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Towards trustworthy lithium-ion battery remaining useful life prediction with uncertainty quantification

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ABSTRACT

With the rapid growth of electric vehicles and renewable energy storage systems, lithium-ion batteries have become a cornerstone of modern energy infrastructure, making accurate and trustworthy remaining useful life (RUL) prediction crucial for operational safety. However, most existing data-driven models predominantly focus on deterministic point estimates while overlooking the underlying prediction uncertainty, which severely undermines the trustworthiness of RUL predictions in safety-critical applications. This paper proposes a Sparsely-gated Mixture of Experts Long Short-Term Memory (SMoE-LSTM) model for trustworthy RUL prediction. The model integrates a sparsely gated mixture of experts with LSTM networks to adaptively capture complex, multi-stage degradation dynamics. The model further incorporates deep evidential regression. This allows for the simultaneous output of RUL estimates and decoupled uncertainties in a single forward propagation. Experimental validation on the MIT-Stanford dataset demonstrates that the proposed method achieves an average RMSE of 16.11 and an R^2 of 0.9821, outperforming baseline methods in both accuracy and uncertainty calibration. Moreover, its single-pass inference design provides substantially higher computational efficiency than MC Dropout and ensemble methods, making it more suitable for real-time battery management systems.

1. Introduction

Amid global efforts to achieve carbon neutrality and transform energy systems, renewable energy-dominated systems and low-carbon transportation are experiencing unprecedented growth [1,2]. Lithium-ion batteries have become a key energy storage solution supporting this transformation due to their high energy density, long cycle life, and environmental friendliness [3,4]. These batteries are widely used in core energy systems of advanced facilities, such as energy storage power stations [5], electric vehicles [6], electric vertical take-off and landing aircraft [7], and aerospace systems [8]. However, the stochastic nature of battery degradation and complex operating conditions render batteries susceptible to safety risks, such as capacity fade and thermal runaway [9–11]. These uncertainties pose severe challenges to the trustworthiness of battery management systems (BMS). Therefore, remaining useful life (RUL) prediction methodologies must evolve from seeking mere accuracy to achieving quantifiable trustworthiness. A trustworthy RUL prediction requires not only precise estimations but also the capability to alert users to potential risks through predictive uncertainty, thereby enabling RUL prediction-based maintenance strategies.

Existing methods for predicting the RUL of lithium-ion batteries are generally divided into two categories [12]: model-driven and data-driven approaches. Model-driven methods establish physical models to describe the battery degradation process [13], with the goal of constructing relationships between observable parameters and aging indicators. These can be further classified into empirical models and physical models. Empirical models are based on mathematical representations of capacity fade trajectories, such as polynomial, double-exponential, and logarithmic models [14–16]. Through data fitting, they map cycle counts to maximum available capacity, enabling RUL estimation and projection of future degradation trends. Physical models capture complex electrochemical reactions and aging mechanisms, making them suitable for adaptive modeling under varying operating conditions [17–19]. However, model-driven approaches depend on accurate physicochemical parameters, which are often difficult to obtain, and typically involve complex equations and high computational costs. In contrast, data-driven methods can capture complex nonlinear degradation patterns directly from historical data without relying on detailed physicochemical parameters [20–22]. Typical approaches

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include autoregressive models, Multi-Layer Perceptrons (MLP), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN), which model the temporal relationship between input variables and RUL. Park et al. [23] proposed a hybrid deep learning model that integrates LSTM attention mechanisms with MLP to improve prediction accuracy by focusing on both statistical and domain-specific features. Chen et al. [24] developed TOP-Net, a parallel hybrid network combining CNN and LSTM, to achieve accurate early-cycle life and RUL prediction for lithium-ion batteries under diverse operating conditions.

Among these methods, the Long Short-Term Memory (LSTM) network has been adopted extensively in the field of battery RUL prediction, owing to its capacity to capture long-term dependencies in capacity fade [25–27]. Zraibi et al. [28] compared DNN, RNN, GRU, and LSTM models, finding that LSTM achieved superior performance in battery RUL prediction with lower prediction errors relative to the other methods. To improve model robustness and adaptability, the Mixture of Experts (MoE) framework has emerged as a promising solution [29–31]. Through a gating mechanism, MoE dynamically assigns different experts to different degradation patterns, which helps reduce modeling bias caused by heterogeneous aging trajectories. Chen and Zhou [32] proposed an AttMoE network that integrates attention mechanisms with a MoE layer, where multiple specialized experts collaboratively enhance the model's ability to capture diverse degradation patterns, thereby improving the accuracy of RUL prediction. Dong et al. [33] proposed LB-DeformNet, which incorporates a Load-Balancing Restriction MoE to ensure uniform expert activation, thereby enhancing model capacity and representation diversity for more robust lithium-ion battery RUL prediction. The integration of diverse experts enables the capture of a wider spectrum of patterns in the data, resulting in enhanced overall performance and more comprehensive representations.

However, conventional data-driven methodologies frequently yield deterministic point estimates of RUL, whereas real-world battery operation is characterized by variable conditions and intricate electrochemical processes. As intuitively illustrated in Fig. 1(a), battery degradation is a highly non-linear process where prediction errors are not static; instead, they exhibit a cumulative propagation effect due to the recursive nature of RUL estimation. As the aging process evolves, these accumulated errors lead to diverging degradation trajectories, making a single point estimate inherently insufficient and potentially misleading. Fig. 1(b) highlights the critical necessity for trustworthy RUL prediction. In the context of BMS, relying solely on point estimates may lead to misguided maintenance decisions and increased costs. Therefore, the provision of RUL predictions accompanied by Uncertainty Quantification (UQ) is essential [34]. A trustworthy RUL prediction should not only provide an accurate estimate but also deliver well-calibrated quantification of its own reliability. This enables decision-makers to understand the limitations and risks associated with the prediction, thereby supporting well-informed judgments in maintenance planning and system management.

Therefore, establishing a trustworthy RUL prediction necessitates advanced UQ capabilities, specifically the ability to decompose uncertainty into its constituent sources. The classification of uncertainty is typically based on its source, with epistemic uncertainty and aleatoric uncertainty being the two primary categories [35]. Epistemic uncertainty, which stems from a lack of knowledge within the model, can be attributed to factors such as the lack of training data or the improper configuration of the model architecture. This uncertainty can be mitigated through the augmentation of datasets or the refinement of the model itself. Conversely, aleatoric uncertainty captures the inherent randomness or noise present in the data, which remains unavoidable even when employing a perfect model. For a trustworthy prediction, the model should distinguish between these two uncertainty sources. High aleatoric uncertainty may indicate noisy operating conditions and the need for signal filtering. High epistemic uncertainty may indicate that the battery is operating in a state that is insufficiently represented in the training data, suggesting the need for more conservative control strategies. This decomposition provides the reason behind an unreliable prediction, fundamentally enhancing the transparency and trustworthiness of the RUL prediction.

UQ in lithium-ion battery RUL prediction currently relies primarily on three methodological approaches: Bayesian neural networks, Gaussian Process Regression (GPR), and ensemble methods [36]. Bayesian neural networks characterize epistemic uncertainty at the model parameter level by imposing prior distributions on network weights and transforming them into posterior distributions [37,38]. Zhang et al. [39] proposed a Bayesian hybrid neural network framework that uses variational Bayesian inference to output predictive distributions. This improves the accuracy and stability of RUL prediction and provides uncertainty estimates. Ke et al. [40] developed a Bayesian neural network using Monte Carlo sampling and variational inference to account for prediction uncertainty. Gaussian process regression is a non-parametric Bayesian method that defines priors directly in the function space and derives complete posterior predictive distributions via covariance kernel functions. This method naturally offers built-in UQ. Liu et al. [41] proposed an LSTM+GPR hybrid framework that uses the empirical mode decomposition to split battery capacity data into some Intrinsic Mode Functions (IMFs) and a residual, LSTM to fit the residual for long-term dependence, and GPR to capture IMFs with uncertainty. Ensemble methods reduce variance and bias at the system level by constructing and combining predictions from multiple heterogeneous base learners [42,43]. Epistemic uncertainty is estimated by evaluating disagreement among the models. Hu and Wang [44] proposed a two-stage ensemble training approach that incorporates a cosine annealing strategy to improve stochastic weight averaging, along with a quantile regression method to quantify uncertainty in RUL prediction, thereby mitigating the overconfidence issue often associated with ensemble methods.

While UQ methods for battery RUL prediction have been preliminary investigated, existing approaches exhibit notable limitations regarding practical trustworthiness. Bayesian methods incur substantial computational overhead due to their reliance on repeated sampling, and their theoretical complexity often hinders the effective decomposition of uncertainty in practice. Gaussian process regression is constrained by its kernel design, which limits scalability, complicates the handling of high-dimensional battery data, and prevents end-to-end feature learning [45]. Although ensemble methods can improve prediction robustness, they cannot explicitly distinguish between different sources of uncertainty, and their multi-model architecture leads to high costs in both training and inference [46]. In contrast, Deep evidential regression offers a paradigm shift towards efficient trustworthiness. As compared in Fig. 1(c), unlike Bayesian or Ensemble approaches that require complex sampling or multiple model runs, deep evidential regression models the higher-order evidential distribution to simultaneously output RUL prediction and decoupled uncertainty estimates in a single forward pass [47,48]. This characteristic perfectly satisfies the real-time performance and trustworthiness requirements in RUL prediction.

In summary, despite advancements in data-driven methodologies and UQ techniques, a critical research gap persists in achieving a synergistic balance among model adaptiveness, computational efficiency, and uncertainty interpretability. Existing UQ frameworks primarily rely on static network architectures that struggle to capture the highly heterogeneous and multi-stage nature of battery degradation. Meanwhile, traditional UQ approaches impose prohibitive computational burdens due to repeated sampling or ensemble execution, and frequently fail to explicitly decouple aleatoric and epistemic uncertainties [49]. Moreover, most existing methods operate as black box models with limited explainability. They provide predictions without revealing why certain decisions are made or what factors contribute to the model's confidence. This lack of transparency further undermines trustworthiness,

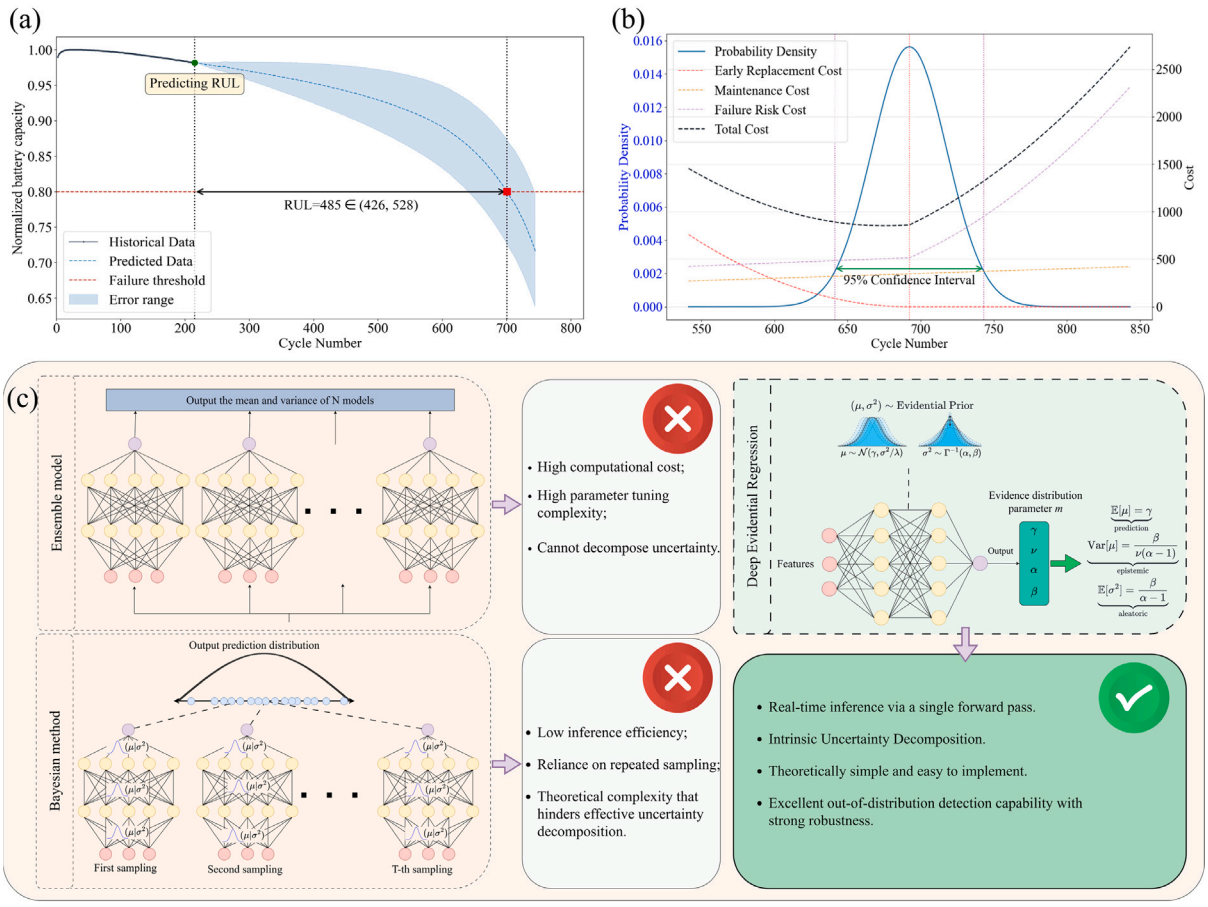


Fig. 1. Conceptual framework illustrating the necessity of trustworthy RUL prediction. (a) Illustration of the cumulative error propagation effect caused by nonlinear battery degradation. As prediction horizons extend, deterministic point estimates fail to capture the naturally diverging trajectories, whereas probabilistic bounds visually reflect this accumulating uncertainty. (b) Uncertainty quantification aids in risk-aware maintenance cost analysis. (c) Methodological comparison demonstrating the superiority of deep evidential regression in terms of inference efficiency and uncertainty decoupling compared to Ensemble and Bayesian approaches.

particularly in safety-critical battery management applications [50]. Therefore, there remains an urgent need for a unified and trustworthy framework for battery RUL prediction. Such a framework should adapt to complex aging mechanisms, provide instantaneous and sampling-free uncertainty quantification, explicitly decouple aleatoric and epistemic uncertainties, and offer interpretable insights into its decision-making process.

This paper proposes a Sparsely-gated Mixture of Experts Long Short-Term Memory (SMoE-LSTM) model for lithium-ion battery RUL prediction with UQ. The proposed framework also improves model interpretability from two aspects. On the structural side, the sparse gating mechanism makes it possible to examine which experts are activated under different degradation conditions. On the predictive side, the explicit decomposition of aleatoric and epistemic uncertainties helps explain whether low-confidence predictions are mainly associated with data noise or limited model knowledge. The primary contributions of this study are as follows:

1. A collaborative SMoE-LSTM framework is constructed to robustly model complex, multi-stage degradation dynamics. By adaptively fusing expert knowledge, this architecture enhances the model's generalization capability across diverse operating conditions, ensuring prediction reliability
2. A trustworthy UQ mechanism based on deep evidential regression is introduced. By employing the Normal-Inverse-Gamma (NIG) prior, the proposed method achieves explicit decomposition of aleatoric and epistemic uncertainties, providing well-calibrated confidence intervals without computationally expensive sampling.
3. Comprehensive validation on the MIT-Stanford dataset demonstrates that the proposed method offers a superior trade-off between accuracy and efficiency. It significantly improves uncertainty calibration compared to baseline methods, providing a verifiable and transparent solution for trustworthy battery RUL prediction.

The remainder of this paper is organized as follows: Section 2 elaborates on the relevant theoretical foundations. Section 3 details the proposed methodology. Section 4 describes the experimental setup and presents an analysis and discussion of the experimental results. Section 5 presents the conclusions of the study.

2. Preliminaries

2.1. Long short-term memory network

The LSTM network is a variant of recurrent neural networks that effectively regulates the accumulation and transfer of information through its gating mechanisms [51]. The LSTM gating mechanism addresses the vanishing gradient issues common in traditional RNN when processing long sequential data. This architecture has been demonstrated to exhibit particular strengths in terms of its ability to

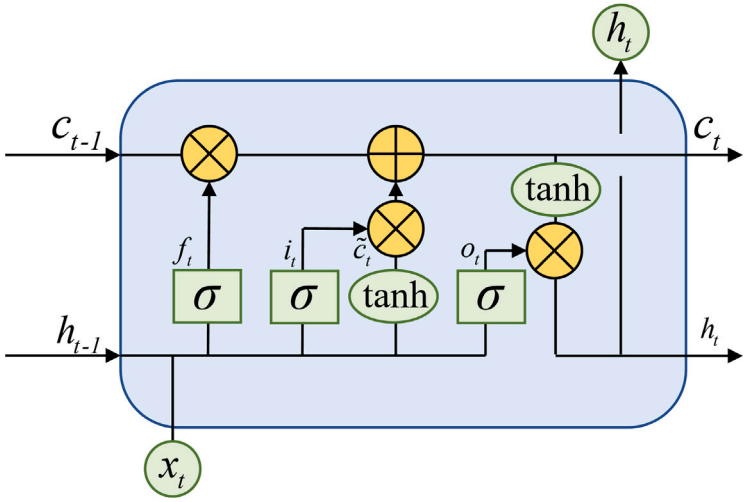


Fig. 2. Basic LSTM unit.

capture long-range dependencies within time series, as illustrated by its detailed structure in Fig. 2.

The core mechanism of LSTM lies in its cell state, where three gating structures work collaboratively to determine: the extent to which information from the previous state is forgotten; the degree to which new information at the current timestep is incorporated; and the information ultimately output to the hidden state. In consideration of the input x_t and the preceding hidden state h_{t-1}, an LSTM unit updates its cell state C_t, consequently yielding the present hidden state h_t through the implementation of the ensuing gating operations. The computational procedure can be summarized as follows:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{2}$$

$$\widetilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{3}$$

$$C_t = f_t * C_{t-1} + i_t * \widetilde{C}_t \tag{4}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{5}$$

$$h_t = o_t * \tanh\left(C_t\right) \tag{6}$$

where f_t, i_t, and o_t represent the activation vectors of the forget gate, input gate, and output gate, respectively; σ and *tanh* denote the sigmoid and hyperbolic tangent activation functions; W and b are the corresponding weight matrices and bias terms; and $*$ indicates the element-wise multiplication operation.

2.2. Deep evidential regression theory

Deep evidential regression is an emerging approach for UQ, which constructs higher-order probabilistic models by introducing evidence priors to provide calibrated and decomposable uncertainty estimates for regression tasks [52]. The fundamental premise of this approach is to circumvent the conventional approach of directly estimating the mean and variance of the target variable. Instead, it employs the establishment of a probabilistic model for the parameters of these likelihood functions, thereby encapsulating both aleatoric and epistemic uncertainty [53].

In the context of a regression problem with a dataset $D = \left\{x_i, y_i\right\}_{i=1}^{N}$, the objective is to learn a functional mapping f, parameterized by a set of weights ω, that provides a reliable approximation to the solution to the following optimization problem:

$$\min_{\omega} J(\omega);\ J(\omega) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_i(\omega) \tag{7}$$

where $\mathcal{L}_i(\bullet)$ denotes the loss function.

Assuming that the observations y_i are drawn independently and identically distributed (i.i.d.) from a Gaussian distribution $y_i \sim \mathcal{N}\left(\mu, \sigma^2\right)$ with unknown mean μ and variance σ^2, the objective is to estimate these parameters probabilistically. Rather than predicting μ and σ^2 directly, deep evidential regression places a higher-order prior distribution over (μ, σ^2). This prior is chosen to be the NIG distribution, which serves as the conjugate prior for the Gaussian likelihood and enables analytical inference, which is defined as follows:

$$p(\mu, \sigma^2 | \gamma, \nu, \alpha, \beta) = \frac{\beta^{\alpha}\sqrt{\nu}}{\Gamma(\alpha)\sqrt{2\pi\sigma^2}}\left(\frac{1}{\sigma^2}\right)^{\alpha+1} \times \exp\left\{-\frac{2\beta + \nu(\gamma-\mu)^2}{2\sigma^2}\right\} \tag{8}$$

where $\Gamma(\bullet)$ denotes the gamma function, and $m = (\gamma, \nu, \alpha, \beta)$ represents the four hyperparameters of the NIG distribution, with $\gamma \in \mathbb{R}$, $\nu > 0$, $\alpha > 1$ and $\beta > 0$. Here, γ indicates the center of the prior mean, ν reflects the confidence in the mean estimate and relates to the number of pseudo-observations, while α and β jointly control the confidence in the variance estimate.

The parameters of this conjugate prior distribution can be interpreted in a way that is consistent with common sense, in that they correspond to pseudo-observations that support specific properties. The total evidence Φ of this evidential distribution, defined as the sum of all inferred pseudo-observation counts $\Phi = 2\nu + \alpha$, reflects the model's overall confidence in the current prediction. By training a neural network to predict m directly from input data, the model learns to express both the predicted target and the associated evidence in a single forward pass, without the need for sampling during inference or out-of-distribution data during training.

This theory shifts uncertainty modeling from the weight space to the likelihood parameter space, offering a scalable and sampling-free alternative for UQ in regression tasks.

3. Methodology

This section delineates a framework for lithium-ion battery RUL prediction. The methodology is structured around three core components to address the challenges of complex degradation dynamics and inherent data variability. First, a rolling sampling strategy is introduced to construct sequential training samples and enable dynamic, cycle-by-cycle RUL estimation. Second, a multi-source feature extraction scheme is detailed, encompassing tailored methods for charging, discharging, and summary data to comprehensively characterize battery health

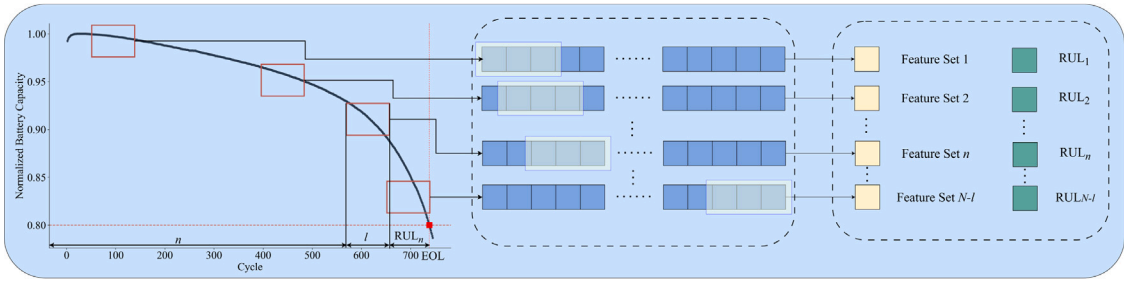


Fig. 3. Schematic of the rolling sampling strategy for RUL prediction.

states. Finally, the core predictive and UQ model is presented, which integrates a collaborative fusion model of SMoE and LSTM for capturing complex temporal patterns with a deep evidential regression layer to jointly output point predictions and well-calibrated epistemic and aleatoric uncertainties in a single forward pass.

3.1. Rolling sampling method for lithium-ion battery RUL

This paper employs a sliding window-based RUL dynamic rolling prediction method, as illustrated in Fig. 3, enabling the prediction to continuously update as the cycle progresses. During the training phase, the complete lifecycle data are utilized. Commencing from the initial cycle, a sliding window of a fixed length l traverses the entire degradation process. Within each designated window, the extraction of features is contingent upon the utilization of data pertaining exclusively to the process variables contained within that particular window. The RUL label is calculated as the difference between the current cycle index and the cycle number at the battery's end-of-life (EOL). The following formula is used to calculate the RUL label:

$$\mathrm{RUL}_n = \mathrm{EOL} - (l + n) \tag{9}$$

where n denotes the number of window shifts. Each shift yields a feature sequence of length l and its corresponding RUL label. For a battery sample with a total cycle life of N, this process can generate $N - l$ training samples. During actual operation, this method enables dynamic rolling output of RUL predictions by continuously collecting process data cycle-by-cycle and feeding the window features into the model. Furthermore, the method does not impose strict requirements on the starting point of the window, allowing it to skip missing cycle samples during data acquisition. This characteristic makes it particularly suitable for application scenarios where initial data may be incomplete or collected intermittently, demonstrating strong data fault tolerance. Overall, this approach provides a feasible pathway for rolling RUL prediction of lithium-ion batteries.

Furthermore, given the raw battery dataset's notable presence of extreme outliers, which have the potential to adversely affect model prediction performance, these anomalous values are eliminated prior to feature extraction based on the statistical 3σ criterion. Missing values are subsequently filled using linear interpolation.

3.2. Feature extraction

To address the issue of intermittent missing data in battery operation and enhance model robustness, this work proposes a multi-source feature extraction strategy. Features are derived from three complementary aspects of the operational data: charging segments, discharging characteristics, and cycle-level summary statistics.

Lithium-ion battery charging protocols vary widely. However, the constant current constant voltage (CC–CV) phase is universally employed to achieve full charge. This study extracts features from key segments within this final CC–CV stage. Specifically, a short voltage segment from the constant current phase and a current segment from the constant voltage phase are selected for analysis, as illustrated in Fig. 4.

Table 1
Features and their meanings.

Feature	Meaning
$V_1, \ldots, V_6$	Resampled voltage data within the $[V_{\mathrm{cutoff}} - 0.2\ \mathrm{V}, V_{\mathrm{cutoff}}]$ interval during the CC charging phase, where V_{cutoff} denotes cut-off voltage.
$I_1, \ldots, I_6$	Resampled current data within the [0.5 A, 0.1 A] range during the CV charging phase.
F_1	Peak value in the IC curve during discharge.
F_2	Peak voltage position in the IC curve during discharge.
F_3	Internal resistance at full charge.
F_4	Maximum discharge capacity of the battery.

Given the high sampling density of raw data, direct use would introduce redundancy. To reduce dimensionality while preserving degradation trends, cubic spline interpolation is applied to resample each extracted sequence into a compact vector of length 6, followed by outlier removal, as illustrated in Fig. 5. The charging features are thus represented as:

$$X_{\mathrm{charge}} = [V_1, \ldots, V_6, I_1, \ldots, I_6] \tag{10}$$

The incremental capacity (IC) curve provides critical insights into battery degradation. Characteristic peaks in the IC curve, resulting from electrode phase transitions, are closely linked to electrochemical state, specifically reflecting the loss of active material (LAM) and the loss of lithium inventory (LLI) during the aging process [54]. As shown in Fig. 6, these peaks attenuate and shift to lower voltages with aging. To quantify this, two key features are extracted from the discharge data: the peak amplitude (F_1) and the corresponding voltage position (F_2). The mathematical representation for the discharge phase feature extraction is formulated as follows:

$$X_{\mathrm{discharge}} = [F_1, F_2] \tag{11}$$

Cycle-level summary data, such as internal resistance and maximum discharge capacity, offer a structured view of battery health. Fig. 7 illustrates their evolution: capacity steadily declines while internal resistance increases, with a strong inverse correlation. A marked acceleration in resistance growth often signals a critical degradation threshold. Therefore, the per-cycle DC internal resistance (F_3), which serves as a macroscopic indicator of solid electrolyte interphase (SEI) growth and impedance rise [55], and maximum discharge capacity (F_4) are extracted as the core features.

$$X_{\mathrm{summary}} = [F_3, F_4] \tag{12}$$

The integration of features derived from charge/discharge cycle data and summary statistics results in the creation of Table 1, thereby facilitating the expression of the complete feature set extracted from the ith cycle as follows:

$$X_i = [V_{i,1}, \ldots, V_{i,6}, I_{i,1}, \ldots, I_{i,6}, F_{i,1}, F_{i,2}, F_{i,3}, F_{i,4}] \tag{13}$$

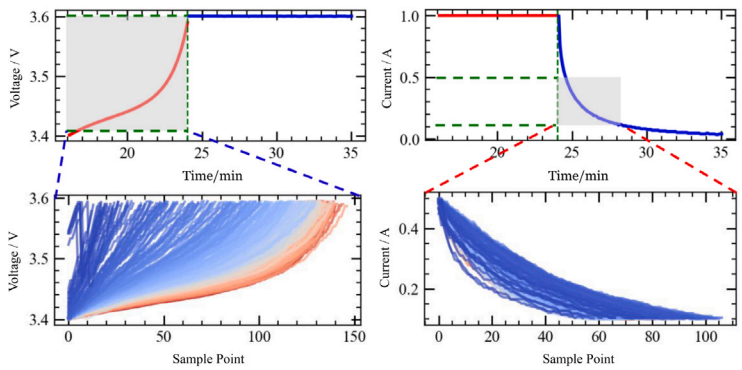


Fig. 4. Capture segment from charging data. Different colors denote batteries with varying cycle lifetimes, with deep red indicating batteries possessing a longer cycle life, while deep blue signifies those with a shorter cycle life.

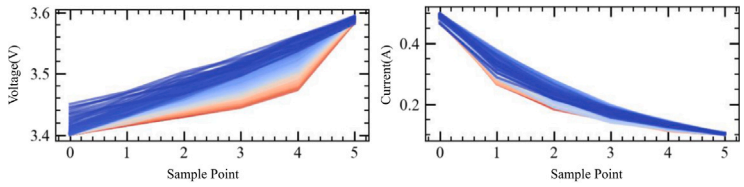


Fig. 5. Interpolated charging segment data.

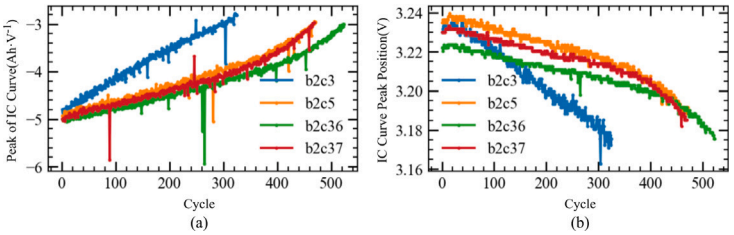


Fig. 6. IC curve peak value and position. (a) depicts the evolution of the peak magnitude in the IC curves with cycling, (b) shows the corresponding shift in the peak voltage position across cycles.

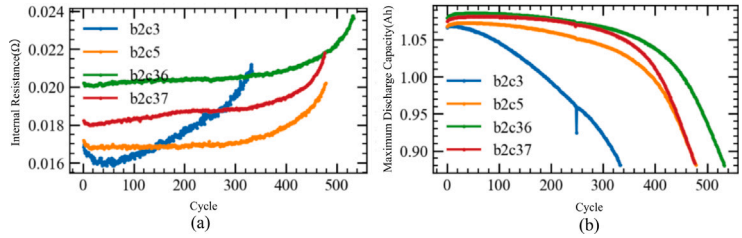


Fig. 7. Evolution of internal resistance and maximum discharge capacity over cycle count. (a) illustrates the evolution curve of internal resistance over cycle count, (b) depicts the variation in maximum discharge capacity across cycles.

Significant discrepancies in scale and unit of measurement are evident among the multi-source features that have been extracted. For instance, characteristic values such as voltage, current, capacity, and resistance are distributed across disparate numerical ranges, while the RUL label predicted by the model, expressed in cycle count, operates on a numerical magnitude substantially higher than other features. In order to enhance the stability, accuracy, and convergence efficiency of model training, this study uses Min–Max normalization to synchronously process all features and labels.

In summary, the selection of this multi-source feature set is not merely driven by empirical data variance, but is deeply grounded in the fundamental electrochemical mechanisms of battery degradation. Specifically, the charging features ($V_{1..6}, I_{1..6}$) reflect changes in polarization and electrochemical impedance during repeated fast cycling. The IC-based features (F_1, F_2) characterize electrode phase-transition behavior and are closely related to LAM and LLI. In addition, the increase in DC internal resistance (F_3) reflects impedance growth, while the maximum discharge capacity (F_4) indicates the overall health state of the battery. Together, these physically meaningful features provide a sound basis for subsequent RUL prediction and uncertainty quantification.

3.3. SMoE-LSTM model

3.3.1. Sparsely-gated mixture of experts

The MoE layer consists of a set of n "expert networks" $E_1, \ldots, E_n$, and a "gating network" G whose output is a sparse n-dimensional vector. Fig. 8 shows an overview of the MoE module. The experts

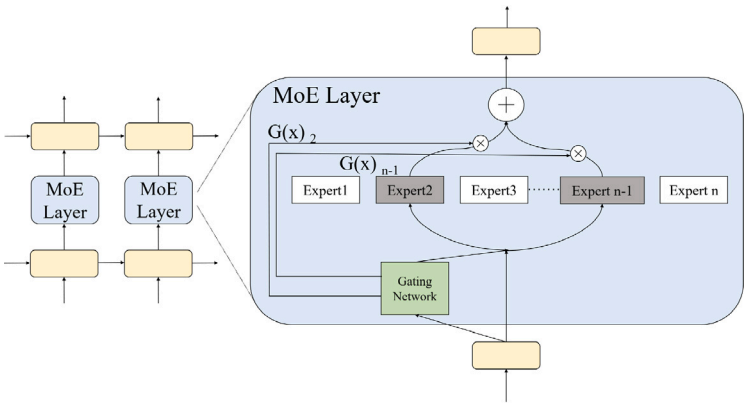


Fig. 8. The structure of SMoE.

are themselves neural networks, each with their own parameters. The output y of the MoE module can be written as follows:

$$y = \sum_{i=1}^{n} G(x)_i E_i(x) \tag{14}$$

where $G(x)$ and $E_i(x)$ denote the output of the gating network and the output of the ith expert network for a given input x.

The gating network is responsible for dynamically calculating the weight allocation of each expert's output in the final prediction based on the input feature x. Its output is a non-negative vector $G(x) = [G(x)_1, G(x)_2, \ldots, G(x)_N]$, where $G(x)_i$ represents the weight of the ith expert. A straightforward non-sparse gating function scheme proposed by Jacob et al. [56] involves multiplying the input by a trainable weight matrix W_g and subsequently applying the Softmax function.

$$G_\sigma(x) = \text{Softmax}(x \cdot W_g) \tag{15}$$

Jacobs' method assigns non-zero weights to all experts. Although it has flexible parameters, it has a large computational cost and lacks sparse control capability. To enhance the sparsity of the model and the expert load balancing capability, this paper uses an improved sparse gating mechanism, termed the noisy Top-K gating mechanism [57]. This method adds noise perturbation on the basis of the standard Softmax gating and retains only the k experts with the strongest responses through the Top-K strategy, while the activations of other experts are forced to be 0, achieving sparse activation. Its overall calculation process is as follows:

$$G(x) = \text{Softmax}(\text{KeepTopK}(H(x), k)) \tag{16}$$

$$H(x)_i = \text{StandardNormal}() \cdot \text{Softplus}((x \cdot W_{\text{noise}})_i) + (x \cdot W_g)_i \tag{17}$$

$$\text{KeepTopK}(u, k)_i = \begin{cases} u_i, & \{j | u_j > u_i\} < k. \\ -\infty, & \text{otherwise.} \end{cases} \tag{18}$$

where W_g and W_{noise} are learnable weights, and the KeepTopK function is used to retain the top-k largest activation values, while the rest are set to $-\infty$.

3.3.2. Overall model structure and evidence layer

To effectively achieve UQ in predicting the RUL of lithium-ion batteries, this paper designs and implements a SMoE-LSTM Model. In contrast to conventional methodologies, such as MC Dropout or model ensemble, SMoE-LSTM Model provides point predictions and epistemic uncertainty, while also explicitly quantifying aleatoric uncertainty.

As demonstrated in Fig. 9, the model is predominantly comprised of a feature extractor and an evidence layer. Initially, the input battery process data is entered into a deep neural network based feature extractor, which results in the acquisition of high-dimensional feature representations. These features are then passed to the evidence layer, which employs fully connected networks to output four parameters $(\gamma, \lambda, \alpha, \beta)$ that collectively define the posterior distribution of the target variable, i.e. the NIG distribution. This distribution provides a point estimate of the RUL and explicitly quantifies the predictive uncertainty.

To enhance the accuracy and robustness of lithium-ion battery RUL prediction, the feature extractor in the proposed model adopts a collaborative fusion model of SMoE and LSTM. This design combines the advantages of time-series modeling and multi-subspace representation, enabling effective capture of long-term dependencies and complex degradation patterns during battery aging. The resulting high dimensional feature representations, which are subsequently passed to the evidence layer, are generated by processing the battery data through this feature extractor.

The mathematical modeling details of the evidence layer are also illustrated in Fig. 9. In the context of the regression task, it is hypothesized that the target values y are drawn from an i.i.d. Gaussian observation model $y_i \sim \mathcal{N}(\mu, \sigma^2)$. In contrast to conventional regression methodologies that predict deterministic means and variances directly, the deep evidential regression approach considers both the mean μ and the variance σ^2 as unknown random variables. The construction of a higher-order posterior distribution $p\left(\mu, \sigma^2 \middle| m\right)$ is achieved through the placement of a conjugate prior, specifically the NIG distribution, over the specified unknowns. The prior is characterized by four evidence parameters $m = (\gamma, \lambda, \alpha, \beta)$, which correspond to the predicted mean, pseudo-count (related to sample size), degrees of freedom, and scale, respectively. Each of these parameters has a clear statistical interpretation.

In summary, the SMoE-LSTM model achieves intrinsic calibration and simultaneous decomposition of uncertainty without relying on external sampling or model ensembles. While maintaining high prediction accuracy, the method effectively enhances the reliability and confidence of lithium-ion battery RUL prediction, making it suitable for practical application in battery management systems. It is noteworthy that, although numerous post-processing-based uncertainty calibration methods exist in current research, deep evidential regression accomplishes intrinsic calibration of predictive uncertainty during model training through an adjustable regularization term, thereby eliminating the need for additional post-processing or retraining.

3.4. Uncertainty quantification mechanism

The UQ mechanism adopted in this study is built upon the higher-order evidential regression theory. This theory constructs a second-order probabilistic model by introducing a NIG evidence prior over the Gaussian likelihood, thereby enabling unified modeling of both aleatoric and epistemic uncertainties.

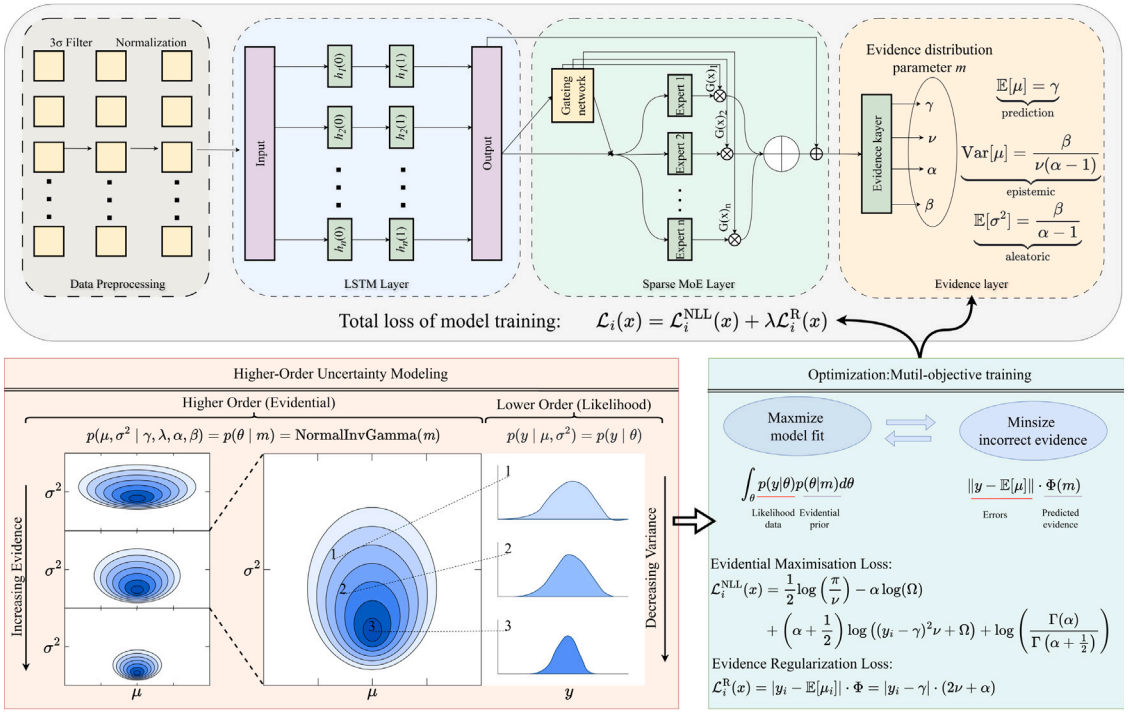


Fig. 9. The structure of SMoE-LSTM model.

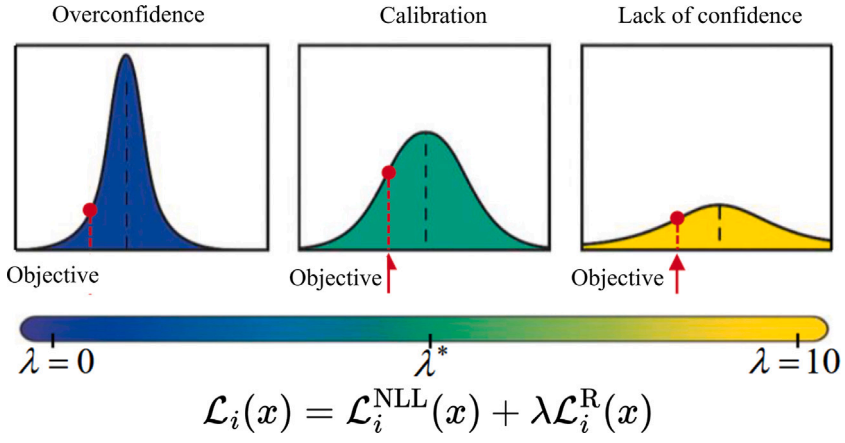


Fig. 10. Tunability of evidential uncertainty.

3.4.1. Uncertainty decomposition

Aleatoric uncertainty captures the variability inherent in observations, even when the same experiment is run repeatedly. Epistemic uncertainty describes the uncertainty in the model's parameter estimates. For the NIG distribution, the predicted value and the aleatoric and epistemic uncertainties can be derived analytically.

$$\underbrace{\mathbb{E}[\mu] = \gamma}_{\text{prediction}} \tag{19}$$

$$\underbrace{\mathbb{E}[\sigma^2] = \frac{\beta}{\alpha - 1}}_{\text{aleatoric}} \tag{20}$$

$$\underbrace{\mathrm{Var}[\mu] = \frac{\beta}{\nu(\alpha - 1)}}_{\text{epistemic}} \tag{21}$$

This decomposition carries distinct physical interpretations. Aleatoric uncertainty characterizes the inherent noise in the data, which cannot be eliminated even with a perfect model. Epistemic uncertainty, on the other hand, reflects the model's knowledge gap due to insufficient training data and can be reduced by collecting more data. It is also important to note that the parameter ν directly modulates the magnitude of epistemic uncertainty.

3.4.2. Learning the evidential distribution

In the context of lithium-ion battery RUL prediction, well-calibrated uncertainty is crucial for ensuring the reliability of model predictions. In this context, calibration signifies the consistency between the model's predictive uncertainty and the observed errors. During training, a joint loss function is employed that is designed to optimize both model fitting and uncertainty calibration simultaneously. In order to achieve intrinsic uncertainty calibration, the loss function is defined as follows:

$$\mathcal{L}_i(\omega) = \mathcal{L}_i^{\mathrm{NLL}}(\omega) + \lambda \mathcal{L}_i^{\mathrm{R}}(\omega) \tag{22}$$

where the parameter λ governs the trade-off between uncertainty inflation and model fitting. Setting λ to 0 results in overconfident estimates, whereas setting it too high leads to overinflation of uncertainty.

$\mathcal{L}_i^{\mathrm{NLL}}(\omega)$ denotes the evidential maximization loss, which aims to align the predicted evidential distribution with the observed data. This is achieved by maximizing the model evidence, i.e. the marginal likelihood. This loss is defined as the negative logarithm of the model evidence.

$$\begin{aligned}\mathcal{L}_i^{\mathrm{NLL}}(\omega) = &\frac{1}{2}\log\left(\frac{\pi}{\nu}\right) - \alpha\log(\Omega) \\ &+ \left(\alpha+\frac{1}{2}\right)\log\left((y_i-\gamma)^2\nu+\Omega\right) \\ &+ \log\left(\frac{\Gamma(\alpha)}{\Gamma\left(\alpha+\frac{1}{2}\right)}\right)\end{aligned} \tag{23}$$

where $\Omega = 2\beta(1+\nu)$. This loss term is designed to encourage the accumulation of evidence in regions where the data provides support.

$\mathcal{L}_i^{\mathrm{R}}(\omega)$ denotes the evidence regularization loss, the purpose of which is to prevent the model from maintaining excessive confidence when predictions are erroneous. The introduction of this regularization term serves to penalize the total evidence $\Phi = 2\nu+\alpha$ when prediction errors are large. The formulation is as follows:

$$\mathcal{L}_i^{\mathrm{R}}(\omega) = \left|y_i - \mathbb{E}[\mu_i]\right| \cdot \Phi = \left|y_i-\gamma\right| \cdot (2\nu+\alpha) \tag{24}$$

The regularization term ensures that when the model's prediction $y_i \approx \gamma$ is accurate, a high level of evidence is maintained without penalty. Conversely, when the prediction error $|y_i-\gamma|$ is large, the model is encouraged to reduce its total evidence $(2\nu+\alpha)$, thereby increasing its predictive uncertainty. In the context of evidence learning, the network $f(x_i;\omega)$ generates four values that correspond to the parameters m. In order to enforce the constraints $\nu > 0$, $\alpha > 1$ and $\beta > 0$, the softplus activation function is employed(with an additional +1 being applied to α to ensure $\alpha > 1$). For $\gamma \in \mathbb{R}$, a linear activation is utilized.

The hyperparameter λ exerts a direct influence on the magnitude of the model's output uncertainty by controlling the regularization strength. As illustrated in Fig. 10, the regularization coefficient λ governs a critical transition in the model's confidence calibration. With increasing λ values, the model progresses from an overconfident regime characterized by insufficient uncertainty estimation to an underconfident state with excessively inflated uncertainty estimates. This progression culminates in an optimal calibration state at a specific value denoted as λ^*, where the model achieves an optimal balance between prediction accuracy and UQ. To establish an equilibrium between these competing objectives across varying λ configurations, this study employed $\lambda = 2$ for all subsequent experimental evaluations.

4. Experiments and results

This section provides a comprehensive evaluation of the proposed SMoE-LSTM model using a public lithium-ion battery dataset. First, the dataset and preprocessing steps are described, followed by the implementation details and hyperparameter settings. The predictive accuracy and robustness of the model are then validated against several common deep-learning baselines. Subsequently, an in-depth analysis of the UQ results is presented, covering both epistemic and aleatoric uncertainties. Finally, the proposed model is benchmarked against established uncertainty-quantification methods with respect to prediction accuracy, computational efficiency, and the quality and calibration of uncertainty estimates. Collectively, these experiments demonstrate the novelty, effectiveness, and practical relevance of the SMoE-LSTM framework for battery management systems.

Table 2
System and software configuration.

Parameter	Detailed configuration
Operating system	Microsoft Windows 10 Professional
CPU	Intel® Core™ i5-10400F CPU @ 2.90 GHz
RAM	16.0 GB
GPU	NVIDIA GeForce GTX 1660 SUPER
VRAM	6 GB
Anaconda3	conda 23.5.0
PyTorch	Torch 2.0.1

4.1. Dataset description

This paper uses a lithium-ion battery dataset that has been collaboratively developed by the Massachusetts Institute of Technology and Stanford University [58]. The dataset under consideration comprises 124 commercial LiFePO4 graphite batteries (APR18650M1A) manufactured by A123 Systems. Fig. 11 shows the schematic diagram of the whole cycle test process of the MIT-Stanford dataset. The batteries were subjected to a series of cycles until they reached the point of failure under conditions of rapid charging. The nominal capacity of the batteries was established at 1.1 Ah, and the nominal voltage at 3.3 V. The EOL point was demarcated as the cycle number at which the state of health declined to 80%. All batteries were tested in a constant temperature environment of 30 °C and subjected to 72 different fast-charging protocols. The charging process adopted a two-stage constant current strategy, denoted as "C1-C2", where C1 and C2 represent the constant current values in the first and second stages, respectively. The charging rates ranged from 3.6 C to 6 C (where 1C = 1.1 A), with 3.6 C being the manufacturer-recommended fast-charging rate. As the state of charge attained 80%, the secondary charging phase concluded. This was succeeded by a 1C CC–CV charging phase, which persisted until the SOC reached 100%. During the charging process, the current gradually decreased to C/50 once the cut-off voltage was attained, at which point the charging cycle terminated. In contrast to the heterogeneity observed in charging strategies, the discharge process exhibited uniformity: all batteries were discharged at a constant current rate of 4C, resulting in a decline to a voltage of 2.0 V. The operating voltage range of the batteries was set between 2.0 V and 3.6 V. The present study utilizes the "2017-06-30" batch of lithium-ion batteries from the dataset that was collectively established by the Massachusetts Institute of Technology and Stanford University. A total of 30 batteries with different charging protocols were randomly selected as experimental subjects, of which 24 were used as the training set and the remaining 6 as the test set, designated as #1, #2, #3, #4, #5, and #6.

We also utilize the XJTU Li-ion battery dataset [59], which comprises two batches with distinct cycling protocols. Batch-1 consists of 8 cells, and Batch-2 consists of 15 cells. For both batches, the first cycle serves as a capacity characterization step: a 0.5 C constant current–constant voltage (CC–CV) charge to 4.2 V with a cutoff current of 0.02 C (40 mA), a 5-min rest, and a 0.2 C (0.4 A) discharge to 2.5 V. In subsequent cycles, Batch-1 cells are charged at 2.0 C (4 A) with a CV cutoff of 0.05 C (0.1 A), rested for 5 min, and discharged at 1.0 C (2 A) to 2.5 V, followed by a 5-min rest. Batch-2 employs a more aggressive 3.0 C (6 A) CC–CV charge with the same 0.05 C cutoff, while the discharge and rest conditions remain identical.

4.2. Configuration details

As shown in Table 2, all experiments were conducted in an Anaconda3-based environment using PyCharm, with PyTorch 2.0.1 as the deep learning framework. The hardware platform consisted of an Intel®Core™i5-10400F CPU, 16 GB RAM, and an NVIDIA GeForce GTX 1660 SUPER GPU with 6 GB VRAM, which was sufficient for training and evaluating the proposed SMoE-LSTM model with deep evidential regression.

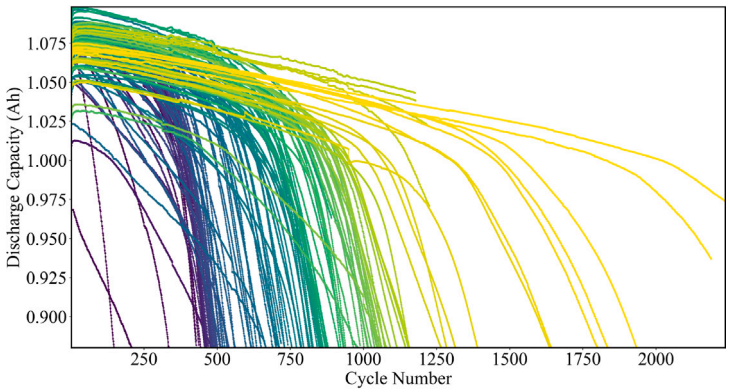


Fig. 11. Battery capacity degradation curve. The color coding system employed to denote the cycle life of the battery is as follows: yellow signifies a longer cycle life, and dark blue indicates a shorter cycle life.

Table 3
Hyperparameter configuration.

Parameter category	Parameter name	Value
Network architecture	Input feature dimension	16
	Hidden layer dimension	128
	Output dimension	1
	Number of experts	10
	Experts per inference step	4
	LSTM layers	2
Training parameters	Batch size	64
	Training epochs	100
	Learning rate	1e−3
	Weight decay	1e−3
	Evidential loss coefficient	2.0
Other parameters	Random seed	42
	Data normalization method	Min-Max normalization
	Optimizer	Adam

The hyperparameter settings are listed in Table 3. These parameters were selected through grid search by considering prediction accuracy, training stability, and computational cost. The input dimension was set according to the extracted battery health features, while the hidden dimension and the number of LSTM layers were chosen to ensure adequate temporal representation capability. In the SMoE-LSTM structure, the number of experts was set to 10 to provide sufficient model capacity for capturing diverse battery degradation patterns under different operating conditions. Meanwhile, 4 experts were activated at each inference step to preserve the sparsity of the mixture-of-experts mechanism. This setting enables the model to benefit from expert diversity without incurring excessive computational cost. It also promotes expert specialization by avoiding simultaneous activation of all experts.

For training, the batch size, learning rate, and epoch number were set to achieve stable convergence. Weight decay was introduced to reduce overfitting, and the evidential loss coefficient was used to balance prediction performance and uncertainty learning. In addition, Min-Max normalization was applied to improve optimization stability, the Adam optimizer was adopted for parameter updating, and the random seed was fixed to ensure reproducibility.

Four metrics were adopted to assess the performance of RUL evaluation methods, which are root mean square error (RMSE), mean absolute percentage error (MAPE), mean absolute error (MAE), and the coefficient of determination (R^2). The mathematical expressions for these metrics are provided below.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{25}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \times 100\% \tag{26}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_i - y_i\right| \tag{27}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{28}$$

where y_i denotes real value, $\hat{y}_i$ denotes predicted value, and n denotes the number of samples.

In order to mitigate the randomness inherent in neural network training, each method underwent 10 independent experimental trials on each test cell. In each trial, the training and test sets were repartitioned, and the model parameters were reinitialized, with the prediction results recorded. In a single trial, the deep evidential regression method required only one training session and one forward pass to output both predictions and corresponding uncertainty estimates. The MC Dropout method necessitated the execution of 30 stochastic forward passes during the prediction phase in order to obtain predictions and uncertainty, while the model ensemble method involved the training of 30 independent models, with predictions and uncertainty derived from their aggregated outputs. To ensure a rigorous and unbiased evaluation of the proposed model's intrinsic effectiveness, all baseline models were carefully configured to maintain a comparable parameter scale and were trained and tested using identical data splits.

4.3. Comparative experimental results of model performance

To comprehensively evaluate the effectiveness and robustness of the proposed SMoE-LSTM under diverse operating conditions, this study compares it with eleven representative baseline models, including conventional neural networks such as CNN and MLP, recurrent architectures such as GRU, and advanced sequence modeling approaches such as Transformer and RUL-Mamba. In addition to these baseline comparisons, an ablation experiment was conducted to explicitly assess the contribution of the sparse mixture of experts module. For this purpose, a degraded variant of the proposed framework without the SMoE component, denoted as Proposed Method (w/o SMoE), was introduced for comparison. The detailed evaluation results of all models on the MIT-Stanford dataset and the two XJTU subsets are summarized in Table 4.

Table 4 presents the performance comparison of different prediction models across the MIT-Stanford dataset and two batches of the XJTU dataset. Overall, the proposed SMoE-LSTM demonstrates superior predictive capability and high robustness. On the MIT-Stanford dataset, the SMoE-LSTM achieves the lowest root mean square error of 16.11, the lowest mean absolute error of 12.77, and the highest coefficient of determination of 0.9821. Although its mean absolute percentage error of 7.42 is marginally higher than the optimal value of 7.11 achieved by the GRU model, it remains highly competitive. The performance

Table 4
Prediction performance comparison.

Model	MIT-Stanford				XJTU-batch1				XJTU-batch2			
	RMSE	MAPE	MAE	R^2	RMSE	MAPE	MAE	R^2	RMSE	MAPE	MAE	R^2
CNN	20.72	9.05	16.54	0.9694	13.38	**7.51**	10.63	0.9866	21.15	10.16	15.50	0.9361
GRU	17.45	**7.11**	13.45	0.9777	25.10	8.02	14.83	0.9528	20.21	11.11	14.26	0.9417
Attention	23.37	8.73	17.80	0.9636	34.21	8.32	20.52	0.9123	19.56	11.98	14.09	0.9454
MLP	21.48	8.94	17.54	0.9680	15.49	8.11	10.82	0.9820	19.78	10.41	13.85	0.9441
Transformer	25.93	9.99	18.02	0.9679	13.18	7.72	**9.52**	0.9870	19.51	10.26	12.28	0.9456
PSO-BPNN	30.66	9.683	24.47	0.8776	43.24	9.76	11.32	0.8600	21.51	10.76	16.21	0.9318
Bi-LSTM [60]	27.07	9.79	18.89	0.9344	26.81	9.94	17.04	0.9462	18.78	10.46	13.52	0.9496
CNN–LSTM [61]	26.31	10.04	18.06	0.9371	27.23	9.60	17.58	0.9445	20.74	8.76	14.74	0.9386
DNN [62]	24.26	9.79	17.69	0.9406	12.11	9.35	10.43	0.9692	19.06	9.69	13.45	0.9481
RUL-Mamba [63]	27.49	9.67	20.69	0.9268	24.09	10.67	12.61	0.9565	19.81	11.71	14.99	0.9421
Proposed method (w/o SMoE)	26.42	9.45	18.31	0.9367	24.84	8.54	15.26	0.9538	21.56	8.78	15.43	0.9336
Proposed method	**16.11**	7.42	**12.77**	**0.9821**	**11.58**	8.63	9.63	**0.9900**	**16.91**	**8.10**	**10.89**	**0.9591**

consistency of the proposed method is further validated on the XJTU datasets, which represent vastly different data distributions and operating conditions. For XJTU-batch1, which consists of 8 cells tested under a 2C discharge rate, the SMoE-LSTM secures the best root mean square error of 11.58 and the highest coefficient of determination of 0.9900. Its mean absolute error of 9.63 is remarkably close to the optimal value of 9.52 produced by the Transformer model. When applied to XJTU-batch2, which involves a larger scale of 15 cells under a more stressful 3C discharge rate, the proposed method completely outperforms all baseline models, sweeping all evaluation metrics with a root mean square error of 16.91, a mean absolute percentage error of 8.10, a mean absolute error of 10.89, and a coefficient of determination of 0.9591. This exceptional performance confirms that the SMoE-LSTM effectively handles diverse datasets and maintains high accuracy under varying discharge conditions and battery sample sizes.

In contrast, several baseline models exhibit noticeable instability or performance degradation across different testing scenarios. For example, PSO-BPNN produces an abnormally high RMSE of 43.24 on XJTU-batch1, suggesting that shallow architectures have limited capability in characterizing the complex and long-term nonlinear degradation behavior of lithium-ion batteries. Similarly, the Attention model shows considerable fluctuation, with its RMSE reaching 34.21 on XJTU-batch1 but decreasing to 19.56 on XJTU-batch2, which indicates insufficient robustness when the data scale and operating conditions change. More importantly, the ablation results provide direct evidence of the effectiveness of the SMoE module. Compared with Proposed Method (w/o SMoE), the complete Proposed Method consistently achieves better results on all three datasets. Specifically, the RMSE decreases from 26.42 to 16.11 on the MIT-Stanford dataset, from 24.84 to 11.58 on XJTU-batch1, and from 21.56 to 16.91 on XJTU-batch2. Similar improvements are also observed in MAE and R^2. These results clearly demonstrate that removing the SMoE module leads to a substantial decline in predictive performance. This performance gap can be attributed to the fact that a single homogeneous temporal modeling architecture is often insufficient to characterize the diverse and nonstationary degradation trajectories of batteries under different operating regimes. By incorporating the sparse mixture of experts mechanism, the proposed model is able to dynamically route heterogeneous degradation patterns to specialized expert subnetworks, thereby reducing feature interference, enhancing representation capacity, and ultimately improving prediction accuracy and cross-condition generalization.

4.4. Analysis of uncertainty quantification

The SMoE-LSTM model requires only a single forward pass to simultaneously generate point predictions of RUL, along with both epistemic and aleatoric uncertainties. This holistic output is fundamental to trustworthy prognostics, as it provides not just a value, but a transparent measure of confidence. More importantly, the model inherently supports explainability from two complementary perspectives.

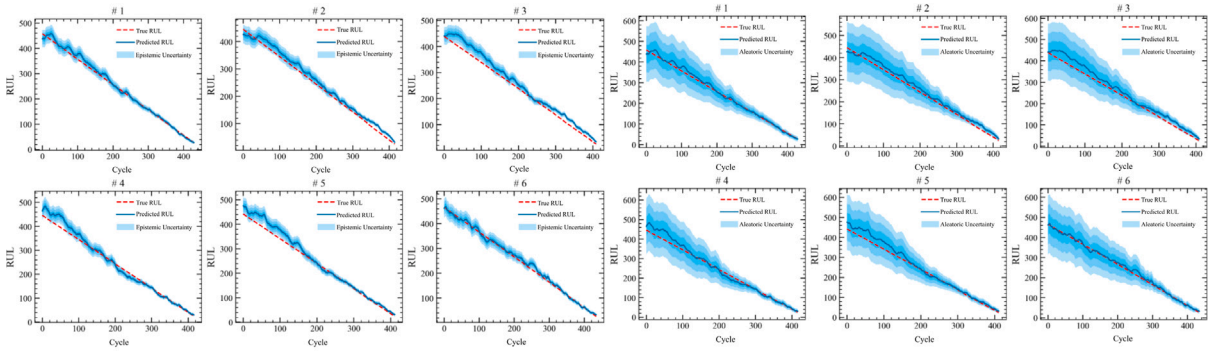
Table 5
Uncertainty quantification results of the SMoE-LSTM model.

Cell	RMSE	Epistemic uncertainty	Aleatoric uncertainty
#1	10.27	5.63	47.76
#2	15.19	5.78	49.35
#3	22.69	5.71	48.41
#4	18.99	5.85	47.89
#5	20.87	5.59	47.06
#6	8.67	6.36	53.16
Average	16.11	5.82	48.94

Structurally, the sparse gating mechanism of SMoE enables researchers to trace which expert sub-networks are activated under different degradation stages, such as early-cycle fluctuations versus late-stage rapid capacity fade, thereby making the model's internal routing behavior transparent. Predictively, the explicit decomposition of aleatoric and epistemic uncertainties offers a clear rationale for why a prediction may be unreliable. High aleatoric uncertainty flags inherent data noise, while high epistemic uncertainty signals that the model is encountering unfamiliar degradation states. The results of the proposed model with regard to UQ were summarized in Table 5. The RMSE across all six test cells was 16.11, establishing a solid baseline for prediction accuracy. Crucially for trustworthiness, the model effectively distinguishes between uncertainty sources. As shown in Table 5, epistemic uncertainty is consistently lower than aleatoric uncertainty. This indicates that the proposed model is indeed capable of successfully learning the underlying degradation patterns of the batteries. The relatively elevated aleatoric uncertainty was indicative of the inherent noise level present in the battery data.

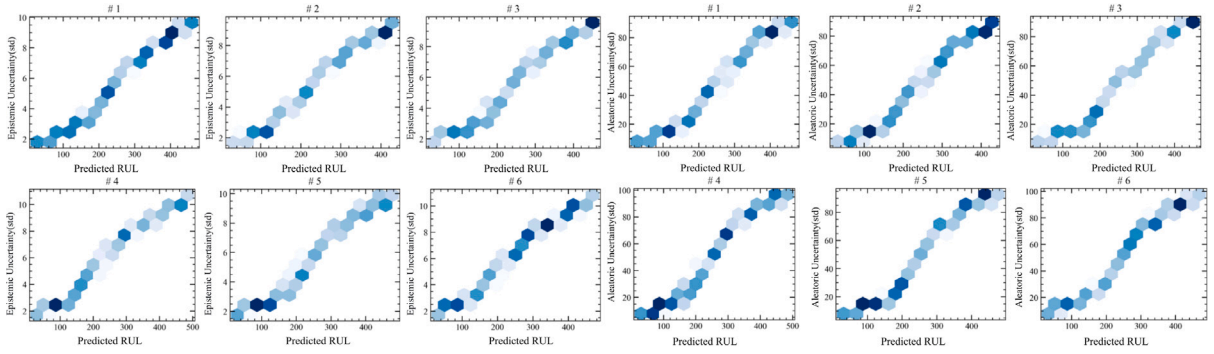
Fig. 12 illustrates the evolution of both uncertainty types during the prediction process. It has been observed that during the early prediction phase (i.e. when far from the EOL), both types of uncertainty remained elevated. This phenomenon occurs because the complex degradation patterns in the initial stages often correlate poorly with the final failure. Concurrently, the impact of inherent data noise was more pronounced, resulting in significantly higher aleatoric uncertainty compared to epistemic uncertainty. This behavior reflects a rational skepticism inherent in a trustworthy model, which acknowledges that initial degradation signs correlate weakly with final failure, thus rightfully withholding high confidence. As cycling progresses and the model accumulates more degradation information, the prediction confidence improves accordingly, leading to a clear downward trend in both uncertainties. This observed behavior aligns well with theoretical expectations, thus validating the reasonableness of the UQ results. This alignment with the physical reality of battery aging confirms that the model's confidence is logically grounded rather than arbitrarily generated.

Fig. 13 demonstrates the relationship between the two types of uncertainty and the predicted RUL, which further corroborates this risk-aware behavior. Uncertainties increase with longer predicted RULs,



(a) Epistemic Uncertainty (b) Aleatoric Uncertainty

Fig. 12. Aleatoric and epistemic uncertainties in RUL prediction. The red dashed line represents the ground-truth RUL, the blue solid line denotes the predicted RUL, and the light-blue shaded area indicates the prediction interval. Subfigure (a) corresponds to epistemic uncertainty, while subfigure (b) corresponds to aleatoric uncertainty. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Epistemic Uncertainty (b)Aleatoric Uncertainty

Fig. 13. Relationship Between Epistemic/Aleatoric Uncertainty and Predicted RUL.

Table 6
Comparative results of prediction accuracy across the three methods.

Cell	MC dropout	Ensemble method	Evidence regression
#1	14.48 ± 0.63	11.23 ± 0.11	**8.45 ± 1.76**
#2	19.63 ± 0.78	14.51 ± 0.14	**13.33 ± 1.98**
#3	20.83 ± 0.93	**14.36 ± 0.18**	23.43 ± 5.57
#4	23.00 ± 0.99	**18.31 ± 0.17**	19.90 ± 3.34
#5	**19.67 ± 1.01**	23.02 ± 0.21	22.14 ± 3.14
#6	22.05 ± 1.16	20.76 ± 0.22	**11.68 ± 3.11**
Average	19.94 ± 2.73	17.03 ± 4.05	**16.49 ± 5.62**

Table 7
Comparative results of inference speed across the three methods.

Cell	MC dropout	Ensemble method	Evidence regression
#1	1.36 ± 0.75	1.21 ± 0.03	**0.04 ± 0.01**
#2	1.21 ± 0.11	1.26 ± 0.03	**0.04 ± 0.01**
#3	1.31 ± 0.15	1.34 ± 0.05	**0.05 ± 0.01**
#4	1.17 ± 0.06	1.24 ± 0.03	**0.04 ± 0.01**
#5	1.20 ± 0.06	1.30 ± 0.07	**0.05 ± 0.01**
#6	1.06 ± 0.05	1.14 ± 0.04	**0.04 ± 0.01**

reflecting the difficulty of long-term forecasting. which provides further support for the aforementioned perspective: predictions involving longer remaining lifespans inherently entail more complex and long-term nonlinear degradation processes, and are more significantly affected by early-stage data noise, consequently leading to higher overall uncertainty. By explicitly quantifying this difficulty, the model alerts the BMS to potential margins of error in long-term planning, thereby enabling safer, more conservative decision-making.

4.5. Benchmark evaluation of uncertainty quantification

While UQ provides the bounds of trust, high prediction accuracy serves as its foundation. The comparative results of prediction accuracy across the three methods were presented in Table 6. With regard to prediction accuracy, the proposed method demonstrated the lowest RMSE in most test cells, including cells #1, #2, and #6, thus indicating its competitive point prediction capability. Despite the fact that the model ensemble approach demonstrated the highest stability, with the smallest standard deviation in RMSE, this advantage came at the cost of training 30 sub-models.

True trustworthiness in RUL prediction implies not just accuracy, but availability and responsiveness. As demonstrated in Table 7, the deep evidential regression method exhibits a substantial advantage in terms of inference speed, outperforming both MC Dropout and model ensemble by one to two orders of magnitude while maintaining an exceptionally low standard deviation. This efficiency is attributed to its single-pass forward propagation design, which eliminates the need for multiple sampling rounds as in MC Dropout or the substantial computational overhead of multi-model ensembles. These characteristics render it particularly well-suited for real-time prediction within RUL prediction.

4.6. Verification of model's calibration

With regard to uncertainty quality, the proposed method demonstrates a clear advantage. Spearman correlation measures the monotonic relationship between the predicted uncertainty and the actual

Table 8
Comparative results of uncertainty quality across the three methods.

Cell	MC dropout	Ensemble method	Evidence regression
#1	0.1989 ± 0.0240	−0.2204 ± 0.1515	**0.3176 ± 0.2146**
#2	−0.1375 ± 0.1041	0.3492 ± 0.1291	**0.5652 ± 0.1877**
#3	−0.4440 ± 0.0631	0.3412 ± 0.2059	**0.4933 ± 0.1672**
#4	−0.4412 ± 0.0521	0.2252 ± 0.1401	**0.6913 ± 0.0885**
#5	−0.2200 ± 0.0515	0.3622 ± 0.1299	**0.6670 ± 0.1345**
#6	−0.3030 ± 0.0526	−0.1838 ± 0.1026	**0.2195 ± 0.1782**

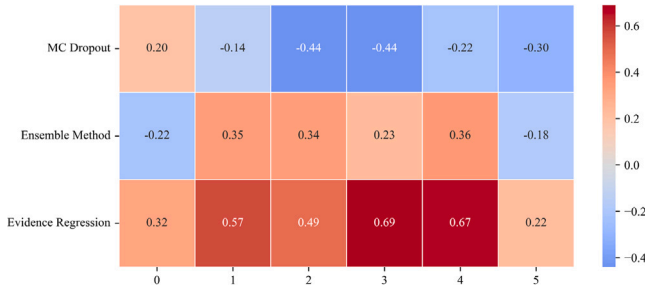


Fig. 14. Relationship between epistemic uncertainty and prediction error.

prediction error. Therefore, a higher positive Spearman correlation indicates better uncertainty calibration, meaning that samples with larger prediction errors are assigned higher uncertainty values. As shown in Table 8, deep evidential regression is the only method that achieves positive correlations across all six cells, with an average Spearman correlation of 0.4923, which is substantially higher than that of the ensemble method (0.1456) and MC Dropout (−0.2245). This result indicates that the proposed method provides a more reliable ranking of prediction confidence over a wide range of samples.

More importantly, the superiority of the proposed method is consistent across different cells rather than being driven by only a few favorable cases. In particular, Cells #4 and #5 exhibit strong positive correlations of 0.6913 and 0.6670, respectively, suggesting that the predicted uncertainty is highly aligned with the actual error distribution in these cases. Even in the most challenging case, Cell #6, the proposed method still maintains a positive correlation (0.2195), whereas both baseline methods show negative values. This observation suggests that evidential regression has better robustness under cell-to-cell variability. In contrast, the ensemble method shows mixed behavior: although it achieves moderate positive correlations in Cells #2–#5, it fails in Cells #1 and #6, indicating limited stability across different data conditions. MC Dropout performs worst overall, producing negative correlations in five out of six cells, which implies that its uncertainty estimates are frequently misaligned with the true prediction errors. From a practical perspective, this improved uncertainty quality is important because it allows unreliable predictions to be more effectively identified, filtered, or prioritized for further inspection. This tendency is also visually supported by Fig. 14, where the proposed method shows a clearer positive association between uncertainty and prediction error than the other two methods.

The hallmark of a truly trustworthy predictor is calibration, which is the alignment between the model's predicted confidence and its actual accuracy. In order to validate the calibration of uncertainty estimates, a further analysis was conducted to ascertain whether samples with high confidence, characterized by low uncertainty, indeed exhibit lower prediction errors. Specifically, the battery samples in the test set were initially sorted in ascending order based on the epistemic uncertainty output by the model, where lower uncertainty indicates higher confidence. In a trustworthy system, selecting samples with higher confidence should strictly guarantee higher accuracy. Subsequently, a range of confidence thresholds were selected, representing the top x% of samples with the highest confidence. The average RMSE of the corresponding samples was then calculated. For instance, a confidence threshold of 0.05 corresponds to calculating the RMSE for the top 5% of samples with the lowest uncertainty.

In principle, if the uncertainty estimation of a method is accurate and well-calibrated, the RMSE should demonstrate a monotonically increasing trend as the confidence threshold expands from 0.05 to 0.95, gradually including more low-confidence samples. This is due to the fact that samples with lower prediction confidence, which corresponds to higher uncertainty, are generally more challenging for the model to predict accurately, thus resulting in larger prediction errors. As demonstrated in Fig. 15, the variation in average RMSE across different confidence quantiles for the six test cells is illustrated under the three methods.

The experimental results demonstrated that the deep evidential regression method exhibited a stable and significant increasing RMSE trend across all test cells, thus confirming its superior calibration characteristics in uncertainty estimation. Conversely, the MC Dropout method demonstrated anomalous behavior in specific cells (e.g., #3 and #4), where the RMSE in high-confidence intervals was unexpectedly higher than in low-confidence intervals. This finding suggests the presence of systematic bias in the estimation of uncertainty, which in turn indicates that the method may, on occasion, result in an overestimation of prediction confidence. This, in turn, gives rise to predictions that appear to be confident but which, in reality, contain a significant degree of error. Whilst the model ensemble method generally displayed an increasing RMSE trend in most cases, the magnitude of change remained limited, reflecting its restricted discriminative capability in uncertainty estimation and its difficulty in effectively distinguishing error differences between high and low-confidence samples.

Further analysis indicates that deep evidential regression performs remarkably well in identifying high-quality predictions. In high confidence intervals (e.g., the top 5% and 25%), this method maintained the lowest RMSE values, thereby demonstrating its capacity to accurately identify the most trustworthy prediction results. This characteristic is of particular value in practical applications: well-calibrated uncertainty estimates can provide battery management systems with a clear decision-making basis. High-confidence predictions can be directly used for maintenance planning, while high-uncertainty predictions should trigger further verification or conservative strategies.

In summary, the superior calibration of the proposed method transforms RUL prediction from a mere numerical output into a trustworthy decision-support tool, enabling it to more reliably identify samples with larger prediction errors and thereby providing trustworthy support for predictive confidence assessment in RUL prediction tasks.

5. Conclusions

Accurate prediction combined with rigorous UQ creates the cornerstone of trustworthy battery management in the energy storage industry. This paper proposes an SMoE-LSTM model, which advances both sequential pattern learning and UQ. The architecture employs a sparse gating mechanism within the collaborative fusion model of SMoE and LSTM to effectively capture temporal dynamics and complex degradation patterns from heterogeneous battery data. Importantly, this sparse gating mechanism provides structural explainability by enabling explicit tracing of expert activation patterns across different degradation stages, revealing how the model adaptively focuses its computational resources. Crucially, the integration of Deep Evidential Regression fundamentally enhances the model's trustworthiness by enabling the simultaneous output of point predictions and decoupled uncertainty estimates. Experimental results on the MIT-Stanford dataset demonstrate the model's effectiveness in achieving trustworthy accuracy. Our model achieves an average RMSE of 16.11 and an R^2 of 0.9821 across six test cells, while providing well-calibrated uncertainty estimates with an average Spearman correlation of 0.49. The single-pass evidential inference further reduces prediction time to 0.04 s per

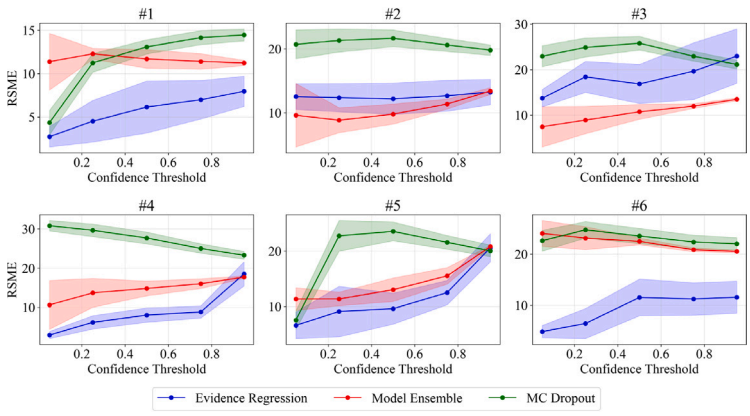


Fig. 15. Comparison of RMSE across different confidence intervals. Curves in different colors represent different methods, with the shaded areas indicating their respective error ranges.

sample, offering a practical and trustworthy solution for real-time BMS applications. By decoupling epistemic and aleatoric uncertainties in a single forward pass, it achieves superior inference efficiency and reliability compared to alternative approaches. This combination of real-time responsiveness and risk awareness offers a practical solution for safety-critical energy storage applications. Future work will focus on three aspects: implementation of the SMoE-LSTM model in battery management system hardware prototypes, validation under more diverse operating conditions and battery chemistries, and integration of electrochemical knowledge to improve early-stage prediction performance.

CRediT authorship contribution statement

Zhibin Zhao: Writing – review & editing, Writing – original draft, Resources, Methodology, Funding acquisition, Conceptualization. **Mingqi Feng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Conceptualization. **Qiuyu Yu:** Resources, Project administration, Methodology, Investigation, Formal analysis. **Hongming Yuan:** Writing – review & editing, Validation, Supervision, Methodology. **Fujin Wang:** Validation, Software, Resources, Formal analysis, Data curation, Conceptualization. **Zhi Zhai:** Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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